

LETTER

Design and analysis of a low voltage overshoot CAN transceiver interface circuit

Weifeng Liu¹, Jinhui Zhou^{1, a)}, Li Zhang^{1, b)}, and Lei Bai²

Abstract To solve the problem pertaining to voltage overshoot arising from frequent switching in CAN transceiver interface circuits, and to mitigate electromagnetic interference while protecting the circuit, this paper proposes a new low-voltage overshooting CAN transceiver interface circuit based on a 0.18 μm BCD. The high-voltage switching transistor is controlled by comparing the bus voltage with a reference voltage to achieve real-time monitoring and protection of the bus voltage. The simulation results show that the interface circuit has excellent power consumption, anti-interference ability, and signal integrity, with a power consumption of 24.2 μA and bus voltage symmetry of 0.934–1.052 under the passive state of the bus.

Keywords: CAN transceiver, voltage overshoot, anti-interference, signal integrity, comparator

Classification: Integrated circuits

1. Introduction

The Controller Area Network (CAN) is a high-reliability serial communication protocol specifically designed for in-vehicle communication. Developed by Bosch in the 1980s, the CAN bus [1] aims to address communication issues between Electronic Control Units (ECU) [2] in automobiles. Due to its high noise immunity, real-time capabilities, and reliability, the CAN bus has been widely adopted in automotive electronics, industrial automation, medical devices, and other embedded systems [3]. A CAN transceiver is a crucial component of the CAN network [3], responsible for converting the micro-controller's digital signals into differential signals on the CAN bus and vice versa, ensuring reliable data transmission in noisy environments [4].

Specifically, the CAN transceiver must perform level shifting, signal driving, and data reception functions [5]. It converts the micro-controller's TTL/CMOS level signals into differential signals for the CAN bus, providing sufficient driving capability to support long-distance communication and multi-node networks [6]. Additionally, it accurately receives and decodes differential signals from the CAN bus. However, traditional CAN transceiver interface circuits typically use diode protection structures [7], which have significant drawbacks such as severe voltage overshoot issues, high power consumption, inadequate noise immunity, and

poor signal integrity.

Moreover, with the introduction of the CAN-FD (Flexible Data-rate) [8] standard, the communication speed of the CAN bus has increased further, and the performance of traditional interface circuits, due to the presence of reverse recovery time, is even more inadequate in high-frequency applications. To address these issues, this paper proposes a novel low-voltage overshoot [9] CAN transceiver interface circuit. This design employs comparators to control the switching of power transistors, achieving real-time monitoring and protection of the bus voltage. This paper will detail the design principles and implementation methods of the new circuit, and validate its performance through simulations, demonstrating its significant advantages in power consumption control, noise immunity, and signal integrity.

2. The proposed interface circuit

Figure 1 illustrates a typical interface circuit schematic, including high-voltage LDMOS and diodes. DriverH and DriverL are slow rate control signals preceding the circuit from the transmitter's intermediate stage. Diodes protect against high voltage but introduce parasitic capacitance and resistance, affecting signal integrity and causing distortion. Under high current, diode dynamic resistance increases voltage drop, leading to higher power consumption. This requires enlarging switch transistors, further increasing power consumption and layout complexity.

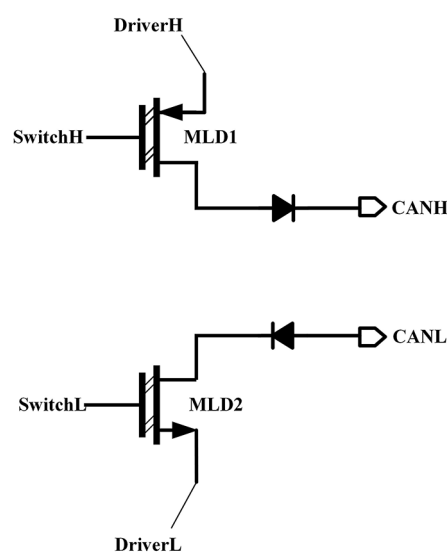


Fig. 1 The typical interface circuit.

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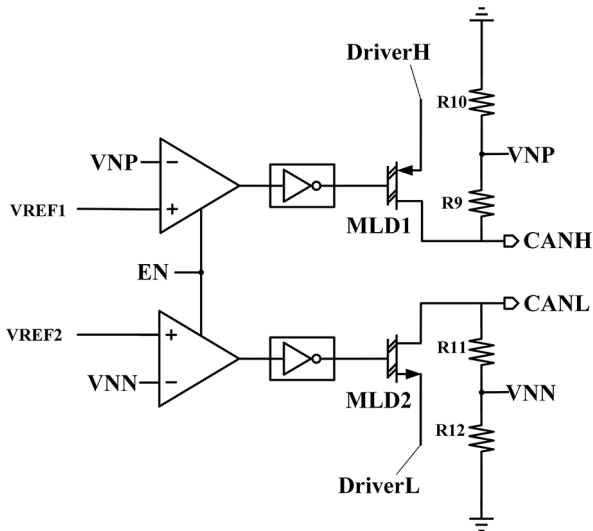


Fig. 2 The proposed interface circuit.

Additionally, the time required for diodes to switch from the conducting state to the reverse blocking state can result in partial current flow during high-speed signal transitions, causing signal distortion and increased noise. In typical interface circuits, the continuous phase-opposite switching signals for MLD1 and MLD2 keep the switch transistors in a constant on-off transition, leading to significant voltage overshoot.

This paper proposes a new interface circuit, as shown in Fig. 2. This design eliminates the diodes and uses a comparator to control the conduction and cutoff of the LDMOS bus switch transistor, protecting and monitoring the bus voltage in real-time by comparing it with reference voltages VREF1 or VREF2. The comparator module consists of two operational amplifiers and an inverter to correct signal logic. The buck module is implemented using a simple resistor divider [10], where R9 and R11 are high-value resistors to withstand abnormal bus voltages and protect the bus, while R10 and R12 are low-value resistors. R9 and R11 are set to the same value, as are R10 and R12, with a ratio of 9:1 between the high and low resistors for easy voltage division calculations. Notably, DriverH and DriverL are slew rate control signals from the intermediate stage of the transmitter, preceding the interface circuit [11, 12]. According to the bus voltage specifications [1] in Table I, the maximum bus voltage for CANH is 4.5 V and the minimum for CANL is 0.5 V. To comply with the CAN bus 11898 protocol standard without hitting extreme values, the protection threshold for CANH is set slightly lower, for example, 3.9 V, and for CANL slightly higher, for example, 1.1 V. In this design, the resistor divider ratio is 1:10, so the reference voltages VREF1 and VREF2 are set to 390 mV and 110 mV, respectively. The working principle is as follows: Under normal conditions, the CAN bus voltage remains safe. The CANH voltage ranges from 2.5 V to 3.9 V, and CANL from 1.1 V to 2.5 V. Operational amplifiers and the inverter maintain MLD1 and MLD2 in the conducting state. Stable signals from the amplifiers are processed by the inverter, ensuring appropriate gate voltages for MLD1 and MLD2. This enables normal current flow and signal transmission on the CAN bus. When CANH voltage

Table I Voltage specifications.

	Recessive			Dominant		
	Min	Typ	Max	Min	Typ	Max
$V_{CANH}(V)$	2	2.5	3	2.75	3.5	4.5
$V_{CANL}(V)$	2	2.5	3	0.5	1.5	2.25
$V_{Diff}(V)$	-0.5	0	0.05	1.5	2	3

exceeds 3.9V, the operational amplifier detects it and signals the inverter. The inverter inverts the signal, raising the gate voltage of MLD1 to turn it off. This protects the transmitter driver module's circuitry from high voltage damage. It stabilizes system operation, reduces current load, and lowers power consumption. The upper operational amplifier compares VNP with VREF1 (390 mV), and the output voltage is expressed as follows:

$$V_{out1} = A \cdot (VNP - VREF1) \quad (1)$$

Where A is the gain of the operational amplifier. When VNP is greater than VREF1, the output Vout1 is at a high level, which, after being inverted by the inverter, remains high, turning off MLD1 and protecting the circuit components from high voltage damage. When the voltage of CANL drops to 1.1 V, the lower operational amplifier detects this change and sends a signal to the inverter. The inverter then inverts this signal, causing the gate voltage of MLD2 to decrease, which turns off MLD2. Specifically, the lower operational amplifier compares VNN with VREF2 (110 mV), and the output voltage is expressed as follows:

$$V_{out2} = A \cdot (VREF2 - VNN) \quad (2)$$

When VNN is less than VREF2, the output Vout2 is at a high level, which, after being inverted by the inverter, becomes a low level, turning off MLD2 to prevent over-current between the ground and the CAN bus, thus protecting the circuit from damage.

For typical interface circuits [13], as shown in Fig. 1, diodes can protect the circuit under high voltage conditions. However, during normal operation, they introduce additional capacitance and resistance, affecting signal integrity and causing signal attenuation or distortion. Furthermore, the voltage drop across the protection diodes increases current, thereby increasing power consumption. This necessitates enlarging the size of the switch transistors to compensate for these losses [14], which raises the complexity and area cost of the layout design. Additionally, during the diode's protection process, the transition time from the conducting state to the reverse blocking state allows a portion of current to pass through. During high-speed signal switching, the diode cannot quickly block the reverse current, leading to signal distortion and increased noise [15–17]. In typical interface circuits, the switch signals of MLD1 and MLD2 are continuous and out of phase, causing the transistors to be in a constant state of switching between on and off, which induces significant overshoot voltage. In high-speed CAN bus applications, frequent signal switching exacerbates this issue.

In the design proposed in this paper, the operational amplifiers in the comparator section can accurately detect voltage changes and quickly output control signals to adjust

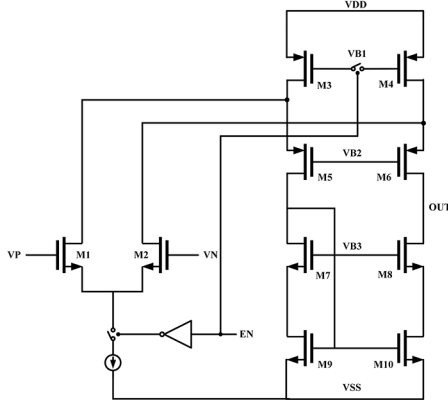


Fig. 3 The amplifier circuit used in this paper.

the state of the switch transistors. Additionally, through the bus monitoring mechanism, the operational amplifiers can continuously monitor voltage changes [18, 19], maintaining the stability and precision of the circuit. This circuit offers superior performance in power consumption control, noise immunity, and signal integrity.

Figure 3 shows the operational amplifier used in the comparator module, which consists of two identical comparators. The main structure of the operational amplifier, comprising M1 to M10, employs a folded cascode configuration [20]. In this structure, the cascode stage provides high output impedance, further enhancing voltage gain compared to a standard common-source stage. VB1 to VB3 are bias voltages provided by the bias circuit. The enable terminal EN is used to control the operational state of the circuit; it stops the operation of the operational amplifier in low-power mode by controlling the current source with an enable signal [20]. The overall design achieves high gain, wide bandwidth, and low-noise signal processing characteristics, while also offering a low-power mode suitable for high-performance analog signal processing applications.

Automotive transceiver chips require low-power, low-temperature-drift circuits to avoid failure. This paper focuses on a simple bandgap reference circuit with these characteristics.

Figure 4 depicts the low-power bandgap reference circuit used in this design. It comprises a startup circuit, core reference part, and feedback loop. Q3 and Q4 form a current mirror pair. In the degenerate state, Q4 charges VREF. Once VREF reaches the required level, Q4 stabilizes and enters cutoff, exiting the startup circuit. The core reference part ensures nearly equal currents in the two branches using M1, M2, M3, and M4. Q1 and Q2 operating at different current densities [21], generate a PTAT voltage ΔV_{BE} , which means they operate under the same collector current but with different collector areas (set to $n:1$, where $Q1:Q2=4:1$ in this case). This voltage is the voltage drop across R3, expressed as:

$$\Delta V_{BE} = V_T \ln(n) \quad (3)$$

And

$$V_{PTAT} = V_T \ln(n) \times \frac{2R_2}{R_1} \quad (4)$$

VREF is the sum of VPTAT and the base-emitter voltage of Q2. It is important to note that in BJTs, there is a cer-

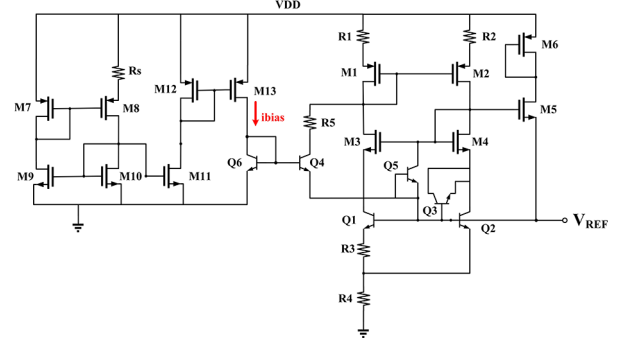


Fig. 4 The bandgap reference circuit used in this paper.

tain amount of leakage current caused by the reverse-biased PN junctions. In this circuit, Q1, Q2, and Q3 experience collector-substrate (CS) leakage. Because the leakage current in reverse-biased PN junctions increases exponentially with temperature [22], its impact becomes significant at high temperatures as it becomes comparable to the quiescent current of the circuit. Considering the effects of leakage current, the main bipolar transistor's CS leakage [23] causes a current discrepancy between the two branches due to the difference in their collector areas. As the temperature rises, this discrepancy is amplified, severely affecting the performance of the reference circuit. In this case, Eq. (4) can be expressed as:

$$V_{PTAT} = V_T \ln \left(n \frac{I_{C1}}{I_{C2}} \right) \times \frac{2R_2}{R_1} \quad (5)$$

On the right side, M5 is mainly used for feedback, ensuring that the total current in the two branches is the same as that in the current mirror of the reference section. At this point, the current in both branches is represented as the sum of the original collector current and the CS leakage current. As shown in Eq. (6):

$$I_{C1} + I_{CS1} = I_{C2} + I_{CS2} \quad (6)$$

Due to the difference in the collector areas of Q1 and Q2, the leakage currents are also different [24], resulting in a discrepancy in the collector currents. This is why neglecting leakage current can lead to errors. To address this, a transistor (Q3) in the cutoff state is introduced, with a size equal to the difference between Q1 and Q2. Its purpose is to provide leakage current compensation [24], ensuring that the CS leakage currents on both sides match, which in turn ensures that the collector currents match, eliminating the effects of non-linearity. Additionally, a single transistor (Q5) located in the central part of the core reference section belongs to the startup circuit, while the total number of Q1, Q2, and Q3 is eight. This arrangement is made for layout matching purposes [25].

The overall circuit of the proposed interface circuit is shown in Fig. 5. The reference voltages VREF1 and VREF2 are provided by a low-power, low-temperature-drift bandgap reference circuit. These reference voltages are then fed into the new interface circuit proposed in this paper for bus voltage detection and protection.

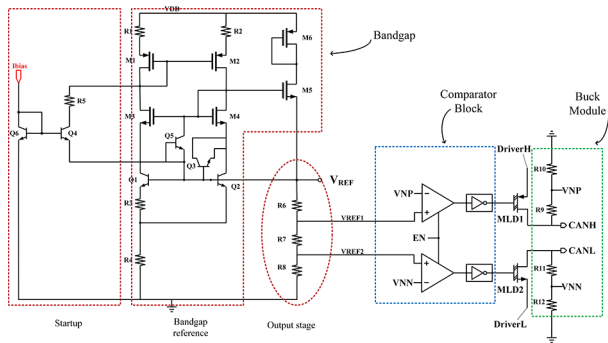


Fig. 5 Schematic diagram of the overall interface circuit.

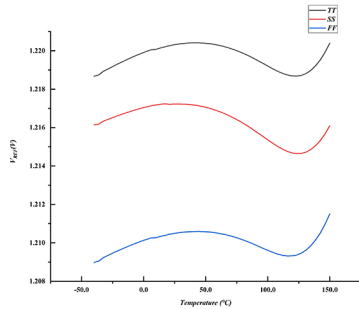


Fig. 6 Temperature characteristics simulation results.

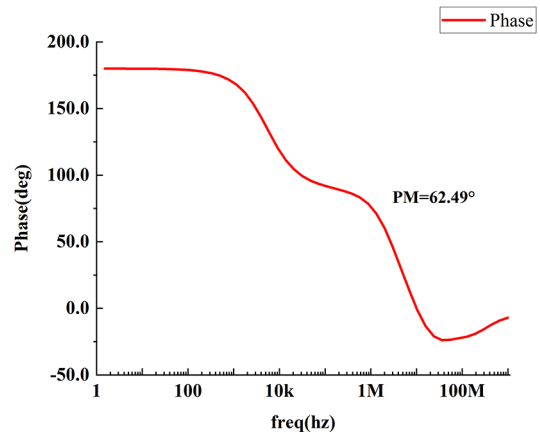


Fig. 7 Simulation results of the loop phase margin of the bandgap.

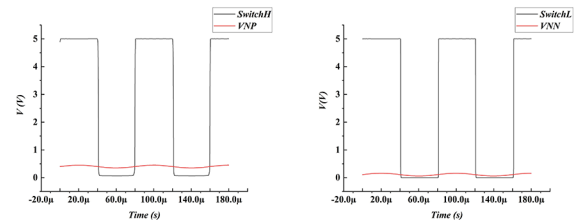


Fig. 8 Simulation of high-side and low-side switch protection logic.

3. Simulation results and analysis

The circuit was designed using a 0.18 μm 40 V BCD process and simulated with Spectre. First, a process corner simulation was performed on the bandgap reference used in this paper. Under a power supply voltage of 5V, the voltage temperature characteristics were simulated for TT (Typical-Typical), SS (Slow-Slow), and FF (Fast-Fast) process corners, with a temperature sweep from -40°C to 150°C . The results are shown in Fig. 6. Within the temperature sweep range, the bandgap reference exhibited good temperature characteristics across all process corners. The voltage difference at the TT process corner was 1.733 mV, with a temperature coefficient of 7.49 ppm/ $^{\circ}\text{C}$. In addition, the feedback loop stability of the circuit was simulated, as shown in Fig. 7. The results indicate a phase margin of 62.49° , demonstrating that the loop is stable and meets the design objectives.

In Fig. 8, the simulation results for the comparator protection logic of the high-side and low-side interface circuits are shown. Each figure has two parts: the lower part shows the divided voltage, and the upper part shows the corresponding switch signal outputs.

For the high-side protection logic, when the divided voltage exceeds 390 mV, the upper switch signal outputs a high level, closing MLD1 to protect the circuit from high voltage. Normally, when the voltage is below 390 mV, the signal is low, keeping MLD1 on to ensure CANH signal transmission.

For the low-side protection logic, when the divided voltage falls below 110 mV, the upper switch signal outputs a low level, closing MLD2 to prevent over-current. Normally, when the voltage is above 110 mV, the signal is high, keeping MLD2 on to ensure CANL signal transmission.

Figure 9 shows the simulation results of the bus voltage

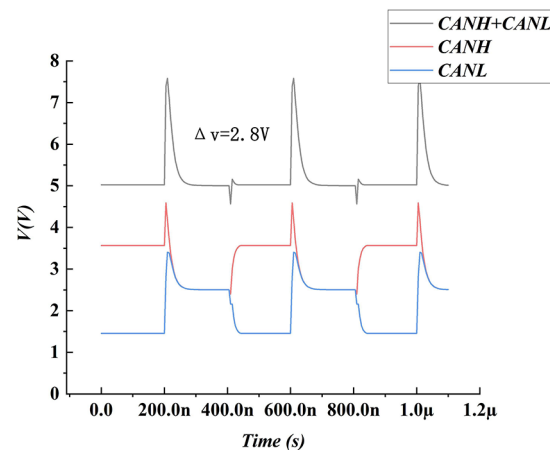


Fig. 9 Bus voltage with the typical interface circuit.

and its symmetry using a typical interface circuit. The black portion at the top of the simulation graph represents the key curve for bus voltage symmetry. As shown in the Eq. (7), symmetry V_{sym} is defined as the ratio of the sum of the bus voltages to the power supply voltage, generally required to be between approximately 0.9 and 1.1. In this paper, the power supply voltage is 5V, meaning that the sum of the bus voltages needs to be between 4.5 V and 5.5 V to achieve good bus symmetry [26–30]. It is evident that in this case, the bus symmetry is poor and has far exceeded the specified range of 4.5 V to 5.5 V. Additionally, there are significant voltage overshoots during bus state transitions, which severely impact the functionality of the transceiver.

Figure 10 shows the simulation results of the bus voltage and symmetry after adopting the interface circuit designed in this paper.

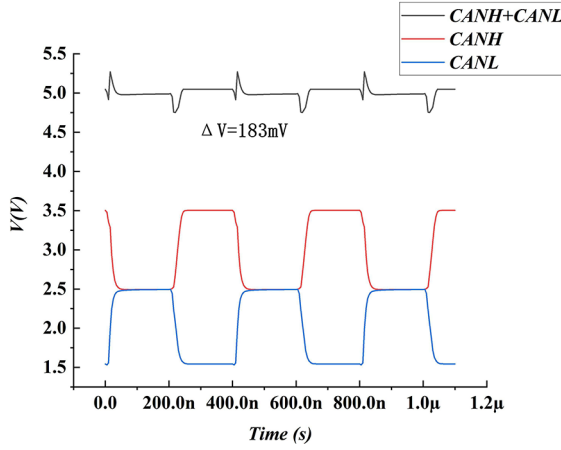


Fig. 10 Bus voltage with the new interface circuit.

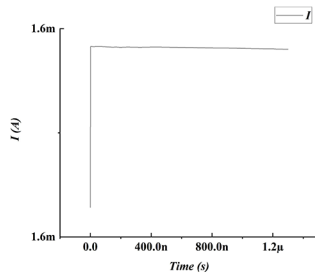


Fig. 11 Current consumption of the typical interface circuit.

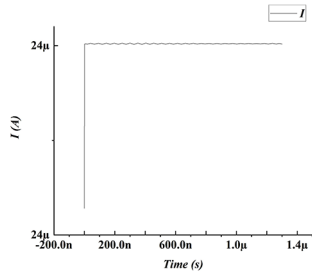


Fig. 12 Current consumption of the new interface circuit.

$$V_{\text{sym}} = \frac{(V_{\text{CANH}} + V_{\text{CANL}})}{V_{\text{DD}}} \quad (7)$$

Figure 11 and Fig. 12 show the results of the average current in recessive state for transmitters using the proposed interface circuit compared to those using a typical interface circuit, highlighting the power consumption.

The power consumption in this design is reduced as the power transistor is not continuously in the conducting state. In the recessive state, the power consumption is only $24.2 \mu\text{A}$, which originates from several sources: $5 \mu\text{A}$ of bias current I_B from the bandgap reference circuit startup, $2 \mu\text{A}$ in the core reference section, approximately $5 \mu\text{A}$ in the output branch, and around $12 \mu\text{A}$ across resistors R_9 to R_{12} . In conventional designs, the power transistor remains in an always-on state, resulting in a continuous path that includes the transistor and leads to a higher power consumption of 1.6 mA .

Table II [31] compares the proposed transceiver with those using conventional interface circuits. By replacing diodes with a comparator-based mechanism, the proposed design

Table II Comparison of the proposed circuit to typical circuit.

	This work	Typ work	[27]	[31]
Tech(μm)	0.18	0.18	0.18	0.18
Operating rate(Mbps)	5	5	5	5
Signal SR(V/ μs)	23.4	21.2	17.4	15.6
V_{sym} (V)	183.2m	2.8	100m	362m
Power consumption(A)	24.2μ	1.6m	1.5m	/

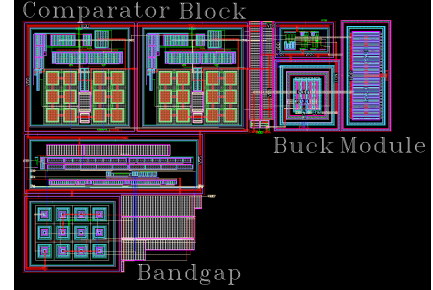


Fig. 13 Interface circuit layout.

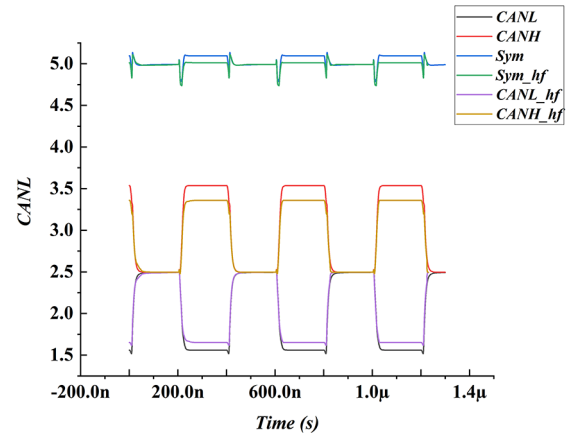


Fig. 14 Post-layout simulation results.

significantly reduces power consumption while preserving bus voltage symmetry. The signal SR (slew rate) is kept under control while avoiding excessive loop delay. The signal rise time is generally maintained within 50 ns , which corresponds to a slew rate of over $20 \text{ V}/\mu\text{s}$.

The layout of the interface circuit is shown in Fig. 13, and the post-simulation results are presented in Fig. 14. The results indicate that the circuit maintains good symmetry and low voltage overshoot during state transitions.

4. Conclusion

By comparing two CAN interface circuits, we conclude that traditional circuits using diode protection suffer from severe voltage overshoot, high power consumption, and poor noise immunity. The new low-voltage overshoot CAN transceiver interface circuit proposed in this paper, designed using a $0.18 \mu\text{m}$ 40 V BCD process, addresses these issues with comparators controlling power transistors for real-time voltage monitoring and protection. Simulation results show the new design excels in power efficiency, noise immunity, and signal integrity, significantly improving stability and reliability. This makes it suitable for high-performance applications, offering significant advantages in reliability and low power

consumption with broad application prospects.

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